Conduction, convection, radiation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify mechanisms of heat transfer and support each classification.

SUCCESS CRITERIA - I can...

1. Describe contact transfer, bulk fluid motion and electromagnetic transfer.
2. Use evidence from the situation.
3. Recognize that more than one mechanism can act.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A device transfers 15 J in 3 s, then performs the same transfer in 6 s. Find both average powers and explain the difference.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $E_i + E_{in} = E_f + E_{out}$.
Account for energy across the system boundary. Conduction: matter interactions; convection: fluid motion; radiation: electromagnetic waves.

Worked reasoning: A metal rod transfers energy from its hot end toward its cool end by conduction. A circulating fluid can carry energy by convection; radiation can cross a vacuum.

Your turn: Draw one distinct, labeled model for each mechanism. Show the direction of energy transfer.

Conduction	Convection	Radiation
Sketch:	Sketch:	Sketch:
_____	_____	_____

Marzano check - circle: ① ② ③

03 Read and interpret the model

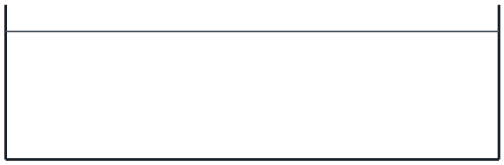
Classify: a metal spoon warming in soup; circulating water in a pot; sunlight warming your hand. Give a reason for each.

Situation	Mechanism	Evidence
Spoon in soup	____	____
Circulating water	____	____
Sunlight on skin	____	____

Marzano check - circle: ① ② ③

04 Apply the physics

A pot of water is heated from below. On the model, draw a circulation loop: warmer, less-dense water rising and cooler water sinking. Explain how this bulk motion differs from energy transfer along a solid metal spoon.



Heater below pot

Marzano check - circle: ① ② ③

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05 Explain your reasoning

An asteroid travels through nearly empty space. Which of these mechanisms can transfer thermal energy to it from sunlight? Why not ordinary air convection?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A rolling object loses 24 J gravitational energy but gains only 16 J translational kinetic energy. Name two possible destinations of the difference.
Show the energy changes and transfers; label units.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

Explain how energy can travel from the Sun to Earth through space.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Design insulation for a hot cup using one change for each transfer mechanism. Explain each change.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3